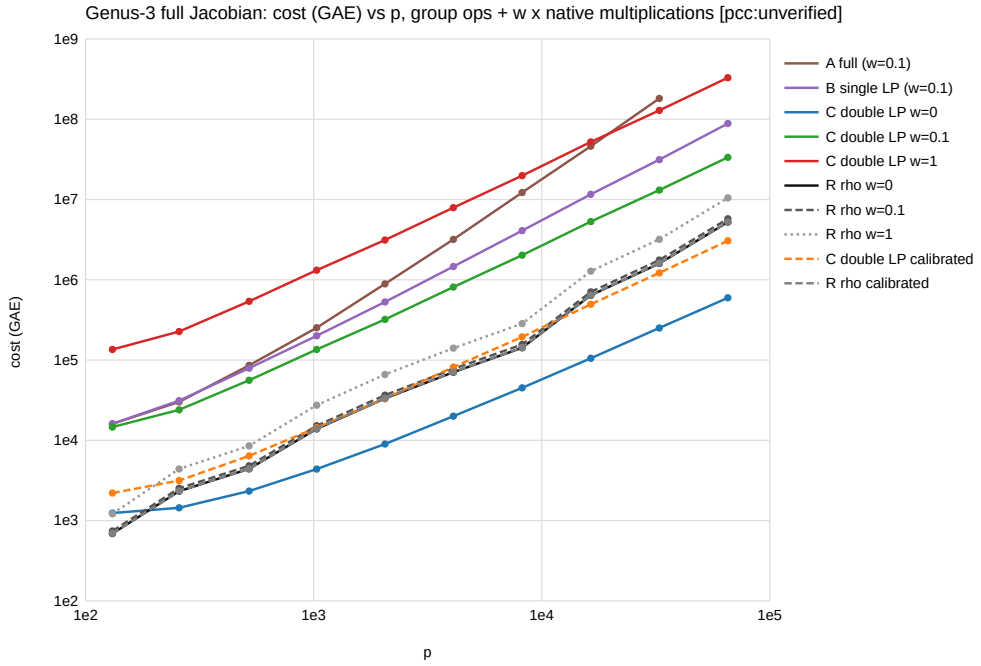
IC positive control: results (2026-11-12)

IC positive control: results (2026-11-12)



figure

Protocol: PROTOCOL.md, frozen by frozen.sha256 before the first measured run. Data: control.json (every run), control.svg, run.log; tables regenerate with `render_results.py control.json` and `uncertainty.py control.json`. Bench record (left open): `benchmarks/records/20261112T2145Z-ic-positive-control.json`. Source: `src/cryptanalysis/chained_transfer/ic_control.rs`, `src/bin/ic_positive_control.rs`. Every numeric statement is `[pcc:unverified]`: no independent validator has replayed it.

What was run

Genus-3 imaginary hyperelliptic curves `y^2 = f(x)`, `deg f = 7`, random monic, over the ten primes `131, 257, 521, 1031, 2053, 4099, 8209, 16411, 32771, 65537` (first prime at or above `2^7 ... 2^16`), on the **full** Jacobian, `#J = h r` with `h <= 64` and `r` prime. Arms: A (basic, every degree-1 place), B (single large prime, `nb = 2 p^{3/4}`), C (double large prime, `nb = 2 p^{2/3}`), R (negation-map rho on the order-`r` subgroup). 572 runs (428 IC, 144 rho): 5 curves per prime up to 2053, 3 at 4099 and 8209, 2 at 16411 and 32771, 1 at 65537, 4 seeds each; A stops at 32771. **All 572 logs certify** (`[k]G == T` by the counted multiplication and by the independent right-to-left multiplication) and no run exhausted its budget [pcc:unverified]. Wall clock for the whole run: 982 s [pcc:unverified]. Mean fraction of trials whose `u` splits completely: 0.155 to 0.177 at every `p` and arm, against `1/6 = 0.1667` for a random cubic [pcc:unverified], so the walk behaves like a random source of divisors.

Verdict against the frozen rule

The instrument cannot detect a win under the AGENTS.md 5a band, within `p <= 65537`. The rule is: C below R at all three prices at some `p`. Result: `strict_all_prices_first_p = null` [pcc:unverified].

- `w = 0`: C is below R at every `p >= 257` (C/R = 0.62 at 257, 0.114 at 65537), above R at 131 (1.80) [pcc:unverified]. At 65537, C spends 5.97e5 group operations against R's 5.24e6 [pcc:unverified].
- `w = 0.1`: C is above R at every `p`; C/R = 5.82 at 65537 (bootstrap 95% interval 3.84 to 9.37), 7.54 at 16411 [pcc:unverified].
- `w = 1`: C is above R at every `p`; C/R = 31.5 at 65537 (interval 20.8 to 52.7), 40.7 at 16411 [pcc:unverified].
- Weaker reading (each price on its own): `w = 0` yes (first `p` = 257), `w = 0.1` no, `w = 1` no [pcc:unverified].
- Order of C and R across the band: it flips between `w = 0` and `w = 0.1`, so by 5a the two are **unresolved**, not a win for C [pcc:unverified].

Informational timing point (not part of the band, never changes the verdict): pricing a native multiplication at its measured cost relative to one Cantor addition (`w_rel` about 0.005 for the relation phase mod `p`, `w_la` about 0.013 for the linear algebra mod `r`; best of 5 on this shared host) gives C/R = 1.15 at 4099, 0.77 at 16411, 0.76 at 32771, 0.58 at 65537 [pcc:unverified]. The bootstrap intervals include 1 at every `p` up to 32771 and exclude it at 65537 (0.38 to 0.94); the first `p` with C/R below 1 (2053, ratio 0.99) is a noisy point (the ratio is back at 1.15 and 1.36 at 4099 and 8209) [pcc:unverified]. A least-squares crossover from the two calibrated fits sits near `p` = 4.7e3 [pcc:unverified]; the data do not support a stable crossing below 16411.

Why C does not win at `w = 0.1` and `w = 1`

The cost is not in the linear algebra and not in the group. At `p = 65537`, C's mean native work is 2.09e8 multiplications in the relation phase and 1.20e8 in the linear algebra, from 5.79e5 trials; the smoothness test (`x^p mod u`, square-and-multiply on `F_p[x]/u`) plus root finding costs about 362 multiplications per trial [pcc:unverified]. A trial costs one group operation, so each group operation drags 362 native multiplications behind it. At a price of 0.1 or 1 GAE per multiplication that is 36 to 362 GAE per trial; rho pays about 1 native operation per step. The asymptotic gap `p^{1/6}` between `p^{4/3}` and `p^{3/2}` is a factor of about 6.4 at `p = 65537` [pcc:unverified], which is the size of the ratios at `w = 0.1` (5.82) and far below those at `w = 1` (31.5). Linear algebra is not the bottleneck either way: Wiedemann on the peeled core (3120 columns at 65537) is 1.20e8 multiplications and 6.61e8 additions, less than the relation phase in multiplications [pcc:unverified].

Per-prime tables (means over all runs of the prime)

Table 1. Counts (mean over runs)

p	runs	A grp	A nat.mul	B grp	B nat.mul	C grp	C rel.mul	
131	5c	1,239	1.480e+05	1,243	1.486e+05	1,242	1.023e+05	
257	5c	1,401	2.875e+05	1,431	2.981e+05	1,443	1.591e+05	
521	5c	2,141	8.367e+05	2,305	7.689e+05	2,337	3.779e+05	1

p	runs	A grp	A nat.mul	B grp	B nat.mul	C grp	C rel.mul	
1031	5c	3,754	2.489e+06	4,428	1.960e+06	4,382	9.069e+05	4
2053	5c	6,814	8.832e+06	9,674	5.194e+06	8,996	2.104e+06	1
4099	3c	13,141	3.170e+07	22,484	1.440e+07	19,959	5.187e+06	2
8209	3c	25,636	1.221e+08	54,874	4.048e+07	45,114	1.300e+07	6
16411	2c	51,074	4.613e+08	1.351e+05	1.147e+08	1.054e+05	3.413e+07	1
32771	2c	1.026e+05	1.813e+09	3.418e+05	3.107e+08	2.513e+05	8.186e+07	4
65537	1c	-	-	8.594e+05	8.751e+08	5.970e+05	2.095e+08	1

Table: total cost in GAE at w = 0, and ratio to R

p	A	B	C	R	A/R	B/R	C/R
131	1,239	1,243	1,242	689	1.80	1.80	1.80
257	1,401	1,431	1,443	2,324	0.60	0.62	0.62
521	2,141	2,305	2,337	4,413	0.49	0.52	0.53
1031	3,754	4,428	4,382	13,884	0.27	0.32	0.32
2053	6,814	9,674	8,996	33,255	0.20	0.29	0.27
4099	13,141	22,484	19,959	70,683	0.19	0.32	0.28
8209	25,636	54,874	45,114	1.424e+05	0.18	0.39	0.32
16411	51,074	1.351e+05	1.054e+05	6.407e+05	0.08	0.21	0.16
32771	1.026e+05	3.418e+05	2.513e+05	1.599e+06	0.06	0.21	0.16
65537	-	8.594e+05	5.970e+05	5.238e+06	-	0.16	0.11

Table: total cost in GAE at w = 0.1, and ratio to R

p	A	B	C	R	A/R	B/R	C/R
131	16,041	16,106	14,656	741	21.64	21.73	19.77
257	30,147	31,244	23,975	2,532	11.91	12.34	9.47
521	85,816	79,191	55,958	4,825	17.79	16.41	11.60
1031	2.526e+05	2.004e+05	1.355e+05	15,234	16.58	13.16	8.89
2053	8.900e+05	5.291e+05	3.208e+05	36,541	24.36	14.48	8.78
4099	3.183e+06	1.463e+06	8.108e+05	77,711	40.97	18.82	10.43

p	A	B	C	R	A/R	B/R	C/R
8209	1.223e+07	4.102e+06	2.024e+06	1.566e+05	78.09	26.19	12.93
16411	4.618e+07	1.160e+07	5.312e+06	7.047e+05	65.53	16.46	7.54
32771	1.814e+08	3.141e+07	1.309e+07	1.759e+06	103.17	17.86	7.44
65537	-	8.837e+07	3.352e+07	5.762e+06	-	15.34	5.82

Table: total cost in GAE at w = 1, and ratio to R

p	A	B	C	R	A/R	B/R	C/R
131	1.493e+05	1.499e+05	1.354e+05	1,217	122.68	123.18	111.27
257	2.889e+05	2.996e+05	2.268e+05	4,398	65.68	68.11	51.56
521	8.389e+05	7.712e+05	5.385e+05	8,533	98.31	90.38	63.12
1031	2.492e+06	1.964e+06	1.315e+06	27,385	91.02	71.73	48.03
2053	8.838e+06	5.203e+06	3.127e+06	66,118	133.68	78.70	47.30
4099	3.172e+07	1.442e+07	7.929e+06	1.410e+05	224.99	102.32	56.24
8209	1.221e+08	4.053e+07	1.984e+07	2.844e+05	429.20	142.49	69.74
16411	4.614e+08	1.148e+08	5.217e+07	1.281e+06	360.17	89.61	40.73
32771	1.813e+09	3.110e+08	1.286e+08	3.197e+06	567.20	97.28	40.24
65537	-	8.760e+08	3.298e+08	1.048e+07	-	83.63	31.49

Table: total cost in GAE at calibrated (informational), and ratio to R

p	A	B	C	R	A/R	B/R	C/R
131	2,400	2,408	2,208	691	3.47	3.48	3.19
257	4,022	4,154	3,151	2,335	1.72	1.78	1.35
521	10,671	9,597	6,397	4,433	2.41	2.16	1.44
1031	31,007	23,126	14,373	13,951	2.22	1.66	1.03
2053	1.125e+05	60,261	33,006	33,416	3.37	1.80	0.99
4099	4.103e+05	1.658e+05	81,757	71,024	5.78	2.33	1.15
8209	1.548e+06	4.490e+05	1.946e+05	1.431e+05	10.82	3.14	1.36
16411	5.855e+06	1.250e+06	4.945e+05	6.437e+05	9.10	1.94	0.77
32771	2.289e+07	3.389e+06	1.221e+06	1.606e+06	14.25	2.11	0.76
65537	-	9.469e+06	3.063e+06	5.262e+06	-	1.80	0.58

Table: sensitivity, native additions included (mean)

p	C native mul	C native mul+add	C total w=1 (mul)	C total w=1 (mul+add)	R total w=1
131	1.341e+05	3.057e+05	1.354e+05	3.069e+05	1,217
257	2.253e+05	5.392e+05	2.268e+05	5.406e+05	4,398
521	5.362e+05	1.338e+06	5.385e+05	1.340e+06	8,533
1031	1.311e+06	3.423e+06	1.315e+06	3.427e+06	27,385
2053	3.118e+06	8.538e+06	3.127e+06	8.547e+06	66,118
4099	7.909e+06	2.321e+07	7.929e+06	2.323e+07	1.410e+05
8209	1.979e+07	6.023e+07	1.984e+07	6.027e+07	2.844e+05

p	C native mul	C native mul+add	C total w=1 (mul)	C total w=1 (mul+add)	R total w=1
16411	5.207e+07	1.661e+08	5.217e+07	1.662e+08	1.281e+06
32771	1.284e+08	4.390e+08	1.286e+08	4.393e+08	3.197e+06
65537	3.292e+08	1.182e+09	3.298e+08	1.182e+09	1.048e+07

Table: calibration (informational)

p	ns / Cantor add	ns / mulmod mod p	ns / mulmod mod r	w_rel	w_la
1031	1040	5.14	14.18	0.0049	0.0136
131	1013	5.13	14.23	0.0051	0.0141
16411	1084	5.02	13.96	0.0046	0.0129
2053	999	4.89	13.51	0.0049	0.0135
257	1030	5.13	14.24	0.0050	0.0138
32771	1081	4.98	13.75	0.0046	0.0127
4099	1062	5.15	14.30	0.0049	0.0135
521	1030	5.13	14.17	0.0050	0.0138
65537	1106	5.04	13.96	0.0046	0.0126
8209	1092	5.15	14.17	0.0047	0.0130

Table: fitted exponents (least squares of ln cost on ln p)

series	slope	coefficient	points p
A_full/group_ops	0.938	5.61	521, 1031, 2053, 4099, 8209, 16411, 32771
A_full/total_cal	1.867	0.0785	521, 1031, 2053, 4099, 8209, 16411, 32771
A_full/total_w0	0.938	5.61	521, 1031, 2053, 4099, 8209, 16411, 32771
A_full/total_w01	1.861	0.653	521, 1031, 2053, 4099, 8209, 16411, 32771
A_full/total_w1	1.866	6.22	521, 1031, 2053, 4099, 8209, 16411, 32771
B_single_lp/group_ops	1.240	0.829	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
B_single_lp/total_cal	1.434	1.12	521, 1031, 2053, 4099,

series	slope	coefficient	points p
			8209, 16411, 32771, 65537
B_single_lp/total_w0	1.240	0.829	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
B_single_lp/total_w01	1.458	8.14	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
B_single_lp/total_w1	1.462	77.9	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
C_double_lp/group_ops	1.158	1.43	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
C_double_lp/total_cal	1.281	1.97	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
C_double_lp/total_w0	1.158	1.43	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
C_double_lp/total_w01	1.325	13.5	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
C_double_lp/total_w1	1.329	128	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
R_rho_negation/group_ops	1.427	0.573	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
R_rho_negation/total_cal	1.427	0.576	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
R_rho_negation/total_w0	1.427	0.573	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
R_rho_negation/total_w01	1.428	0.624	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537
R_rho_negation/total_w1	1.433	1.08	521, 1031, 2053, 4099, 8209, 16411, 32771, 65537

Crossover


```
{
 "0": {
  "first_p_C_below_R": 257,
  "fit_crossover_p": 29.903517315754204
 },
 "0.1": {
  "first_p_C_below_R": null,
  "fit_crossover_p": 10348783237051.592
 },
 "1": {
  "first_p_C_below_R": null,
  "fit_crossover_p": 1.2142291186455482e+20
 },
 "cal": {
  "first_p_C_below_R": 2053,
  "fit_crossover_p": 4706.030054311983
 }
}
```


strict_all_prices_first_p: None all_certified: True runs: 572 ic: 428 rho: 144 splits/trials per arm and p (smoothness sanity): p=131 A_full: 0.1552 p=131 B_single_lp: 0.1550 p=131 C_double_lp: 0.1551 p=257 A_full: 0.1639 p=257 B_single_lp: 0.1633 p=257 C_double_lp: 0.1638 p=521 A_full: 0.1754 p=521 B_single_lp: 0.1770 p=521 C_double_lp: 0.1766 p=1031 A_full: 0.1629 p=1031 B_single_lp: 0.1631 p=1031 C_double_lp: 0.1634 p=2053 A_full: 0.1695 p=2053 B_single_lp: 0.1693 p=2053 C_double_lp: 0.1693 p=4099 A_full: 0.1667 p=4099 B_single_lp: 0.1666 p=4099 C_double_lp: 0.1669 p=8209 A_full: 0.1689 p=8209 B_single_lp: 0.1688 p=8209 C_double_lp: 0.1689 p=16411 A_full: 0.1661 p=16411 B_single_lp: 0.1651 p=16411 C_double_lp: 0.1654 p=32771 A_full: 0.1647 p=32771 B_single_lp: 0.1652 p=32771 C_double_lp: 0.1650 p=65537 B_single_lp: 0.1664 p=65537 C_double_lp: 0.1666

Fitted exponents (least squares of `ln cost` on `ln p`, points `p >= 500`)

Points used are listed in the table above (“points p”). B and C use the eight primes 521 to 65537; A uses 521 to 32771 (seven points); R uses the eight primes [pcc:unverified].

series	exponent (measured)	theory	note
A group ops	0.938	1	`~3 p` trials
A total, `w = 0.1`	1.861	2	linear algebra ~ `p^2`
B group ops	1.240	1.5	`p^{3/4}` base; still pre-asymptotic
C group ops	1.158	4/3	`p^{2/3}` base; still pre-asymptotic
C total, `w = 1`	1.329	4/3	
C total, `w = 0.1`	1.325	4/3	
C total, calibrated	1.281	4/3	

series	exponent (measured)	theory	note
R group ops (and total)	1.427	3/2	cofactor `h` varies with `p`: `sqrt(r)` carries `h^{-1/2}`

The measured slope gap between C and R is 0.10 at `w = 1` and 0.15 at the calibrated point, against the asymptotic 1/6 = 0.167 [pcc:unverified]. Extrapolating the two `w = 0.1` fits gives a crossover near `p` = 1.0e13, and at `w = 1` near 1.2e20 (`crossover` block of `control.json`) [pcc:unverified]. These are extrapolations over many orders of magnitude from a gap of about 0.1 in slope; they say “not in range”, not a prediction.

Prediction scorecard (PROTOCOL.md section 7)

#	Prediction	Outcome	Result
P1	`w = 0`: C below R at every `p >= 257`	C/R from 0.62 (257) to 0.114 (65537), all below 1	held
P2	`w = 0.1`: C/R > 1 everywhere; C/R at 16411 in [8, 40]; smaller at 65537 than at 16411	C/R > 1 everywhere; 7.54 at 16411 (outside the interval, just below 8); 5.82 at 65537 (smaller)	**partly falsified** (interval)
P3	`w = 1`: C/R > 1 everywhere; C/R at 16411 in [40, 250]	40.7 at 16411, above 1 everywhere	held (at the lower edge)
P4	calibrated C/R at 16411 in [0.7, 2.0]	0.77	held
P5	decision rule fails	`strict_all_prices_first_p = null`	held
P6	exponents: C group ops 1.0 to 1.4; C total `w = 1` 1.25 to 1.55; R total `w = 1` 1.35 to 1.6; A total `w = 0.1` 1.8 to 2.15; A group ops 0.95 to 1.1	1.158; 1.329; 1.433; 1.861; 0.938	four held, **A group ops falsified** (0.938, below 0.95)
P7	`w = 0.1` fitted crossover above 1e6	1.0e13	held
P8	every log certifies	572 of 572	held

Two of the eight predictions have a falsified part, each on a margin: the `w = 0.1` ratio at 16411 is 0.46 below its interval (C closer to R than predicted) and the A group-operation slope is 0.012 below its range. Neither changes the verdict [pcc:unverified].

Uncertainty (bootstrap over runs, 2000 resamples, seed 20261112)

Rho’s run-to-run spread dominates: its coefficient of variation is 0.37 to 1.10 per prime, against 0.002 to 0.73 for C (the largest for C at 131) [pcc:unverified]. Mean rho at 65537 rests on 4 runs.

p	n_C	n_R	key	C/R	95% interval	CV(C)	CV(R)
131	20	20	total_w0	1.804	[1.343, 2.580]	0.725	0.365
131	20	20	total_w01	19.769	[14.526, 29.218]	0.779	0.372
131	20	20	total_w1	111.267	[81.246, 162.666]	0.783	0.411
131	20	20	total_cal	3.194	[2.445, 4.423]	0.654	0.365
257	20	20	total_w0	0.621	[0.493, 0.803]	0.042	0.545
257	20	20	total_w01	9.470	[7.530, 12.124]	0.058	0.547
257	20	20	total_w1	51.560	[41.241, 66.320]	0.060	0.561
257	20	20	total_cal	1.350	[1.093, 1.724]	0.058	0.545
521	20	20	total_w0	0.530	[0.400, 0.716]	0.058	0.674
521	20	20	total_w01	11.599	[8.837, 15.853]	0.089	0.677
521	20	20	total_w1	63.115	[47.319, 85.936]	0.090	0.690
521	20	20	total_cal	1.443	[1.100, 1.959]	0.100	0.674
1031	20	20	total_w0	0.316	[0.221, 0.463]	0.043	0.906
1031	20	20	total_w01	8.893	[6.228, 13.035]	0.064	0.908
1031	20	20	total_w1	48.029	[33.357, 71.955]	0.065	0.918
1031	20	20	total_cal	1.030	[0.717, 1.535]	0.071	0.906
2053	20	20	total_w0	0.271	[0.201, 0.397]	0.036	0.781
2053	20	20	total_w01	8.780	[6.562, 12.906]	0.070	0.782
2053	20	20	total_w1	47.299	[34.957, 71.427]	0.071	0.785
2053	20	20	total_cal	0.988	[0.728, 1.461]	0.082	0.781
4099	12	12	total_w0	0.282	[0.171, 0.591]	0.020	1.103
4099	12	12	total_w01	10.434	[6.388, 22.237]	0.019	1.104
4099	12	12	total_w1	56.245	[33.902, 119.934]	0.019	1.106
4099	12	12	total_cal	1.151	[0.694, 2.445]	0.021	1.103

p	n_C	n_R	key	C/R	95% interval	CV(C)	CV(R)
8209	12	12	total_w0	0.317	[0.243, 0.437]	0.016	0.530
8209	12	12	total_w01	12.925	[9.987, 17.740]	0.033	0.530
8209	12	12	total_w1	69.745	[52.736, 96.106]	0.033	0.531
8209	12	12	total_cal	1.360	[1.046, 1.848]	0.042	0.530
16411	8	8	total_w0	0.164	[0.112, 0.277]	0.015	0.690
16411	8	8	total_w01	7.538	[5.222, 12.472]	0.009	0.690
16411	8	8	total_w1	40.729	[27.900, 67.977]	0.009	0.690
16411	8	8	total_cal	0.768	[0.517, 1.262]	0.009	0.690
32771	8	8	total_w0	0.157	[0.105, 0.251]	0.011	0.704
32771	8	8	total_w01	7.444	[4.937, 11.685]	0.005	0.705
32771	8	8	total_w1	40.237	[26.738, 62.560]	0.005	0.705
32771	8	8	total_cal	0.760	[0.503, 1.186]	0.005	0.704
65537	4	4	total_w0	0.114	[0.079, 0.184]	0.007	0.524
65537	4	4	total_w01	5.818	[3.839, 9.370]	0.003	0.524
65537	4	4	total_w1	31.485	[20.782, 52.650]	0.003	0.524
65537	4	4	total_cal	0.582	[0.384, 0.938]	0.002	0.524

What it implies for the earlier “distance to rho” figures

1. **The instrument can read a win at `w = 0`.** In the one regime where IC should beat rho, group operations alone put C at 0.114 of rho at 65537, with a slope gap of 0.27 to rho at `w = 0` (1.158 against 1.427) [pcc:unverified]. So the earlier `w = 0` ratios (for example IC2/rho_E = 47.11 at 64007 in `tower_large`) are real losses of the group-operation count, not an artefact of an instrument that cannot show a win [pcc:unverified].
2. **The instrument cannot read a win at `w = 0.1` or `w = 1` at these sizes, even when the algorithm wins asymptotically.** The best known IC variant in its winning regime sits at 5.8 to 9.4 times rho at `w = 0.1` and 20.8 to 52.7 times at `w = 1` at `p = 65537` (intervals) [pcc:unverified]. The earlier statements “above rho at every weight” in `tower_large` therefore carry no information about the transfer at `w = 0.1` and `w = 1`: the positive control reads the same way. Only the `w = 0` column, and the order across the band, discriminate [pcc:unverified].
3. **Distances read at `w = 0.1` and `w = 1` overstate a gap that a calibrated multiplication price would shrink.** The control’s own C/R drops from 5.82 (`w = 0.1`) and 31.5 (`w = 1`) to 0.58 at the measured price, because the band prices a modular multiplication in `F_p` at 0.1 to 1 Cantor additions while a Cantor addition takes the time of about 80 to 200 of them here (ns ratio 0.013 to 0.005) [pcc:unverified]. Any earlier

distance whose gap is below the control's own (about 6 at `w = 0.1`, about 31 at `w = 1`) is not distinguishable from a winning algorithm's reading [pcc:unverified].
4. **Sparse linear algebra and large primes were not what limited the earlier rows.** The genus-3 linear algebra quoted in NEXT_STEPS (2.2e9 mulmods at 64007, a different instance and algorithm) compares with 1.20e8 mulmods for C and 5.73e8 for B at 65537 here, so a Wiedemann core on large-prime relations is cheap by comparison; it is not a like-for-like figure [pcc:unverified].

The earlier rows are not revised by this record; it supplies the yardstick: a win in the known-winning regime shows up at `w = 0`, is unresolved inside the band at 65537, and needs the calibrated price to cross 1.

Deviations and limits

- Arm A was not run at 65537 (`--full-cap 32771`): its dense core at 65537 was the one unaffordable job. A therefore has seven fit points.
- Single curve and 4 runs at 65537; two curves at 16411 and 32771. The ratios there carry the intervals above, not a point estimate.
- `r` is about `p^{2.3}` to `p^{2.8}`, not `p^3` (cofactor `h` up to 28 in the admitted curves, plus `#J` fluctuation); rho scales with `sqrt(r)`, IC with `p`. R's fitted exponent (1.427) carries `h` noise from curve to curve [pcc:unverified].
- The native-operation count of rho is one canonicalisation per step (never below the step count); a fuller count of rho's table work would raise R and favour C slightly. The relation phase counts only the work the code does (no skipping of zero coefficients, no Legendre pre-filter on the discriminant, which would cut the smoothness test by less than a factor of 2).
- Linear algebra: only the final build, peel and solve are counted; every run solved at the first attempt (`attempts = 1` in all 428 IC runs), so no retries are hidden. The stopping rule used an uncounted incremental check, which assumes its cost is at most the counted union-find operations.
- Primary price counts multiplications (the repository convention). Counting additions too raises C's native work 2.3 to 3.6 times (table above); the order of C and R is unchanged at every band price [pcc:unverified].
- Timing calibration is a best-of-5 micro-benchmark on a shared host; it is an extra point, never part of the verdict.
- The protocol file was written after the code and before any data; disclosed in PROTOCOL.md.

Checks run

`cargo test --release --lib chained_transfer::ic_control`: 4 passed (mulmod against u128, roots and smoothness, instance and arms with relation verification, rho against IC on one instance). `cargo clippy --release --all-targets -- -D warnings`: clean for the whole crate. `python3 tools/pcc/pcc.py lint-prose` on this file and PROTOCOL.md: 0 findings.